1. Introduction to Data Science

Data Science with Applications in Natural Language Processing

Learning goals

By the end of this chapter we will be able to:

- **distinguish** data, information and knowledge
- **classify** data as structured, semi-structured, unstructured
- **audit** a dataset for quality problems
- **recognise** where subjectivity enters
- **describe** the life cycle of a project
- **turn** text into vectors (Bag-of-Words)

Chapter map

1. What are data?
2. Quality and subjectivity
3. Roles and applications
4. The life cycle of a project
5. Natural language processing at the centre

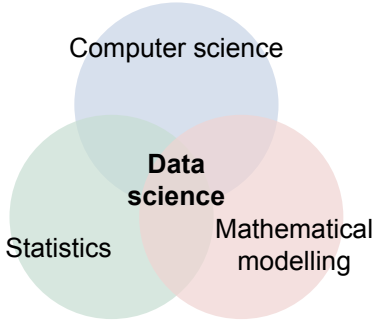
A question that will follow us

What do citizens say about a new policy?

```
posts = [
    "The new policy is unfair",
    "I support the reform",
    "I do not trust the system",
]
```

We will come back to these three posts in every section.

What is data science?



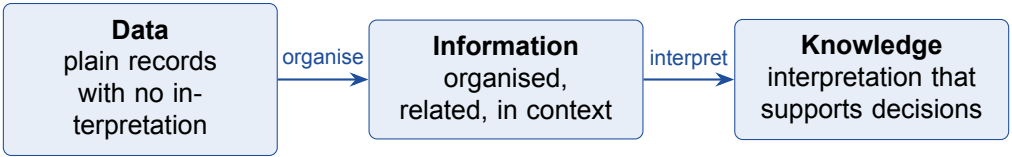
- An interdisciplinary field
- Goal: **knowledge from data**
- Data that are large, heterogeneous, noisy
- Not just description: understanding, prediction, decisions

Recorded observations

of events, objects, states, processes

- The raw material of data science
- On their own: **raw** and **fragmentary**
- They gain meaning inside a frame of interpretation

Data → information → knowledge



Data science is about **exactly this transition**.

Example: three levels, same posts

Data

```
posts = [
  "The new policy
   is unfair",
  "I support the
   reform",
  "I do not trust
   the system"]
```

Information

```
lengths = [len(p.split())
           for p in posts]
# [5, 4, 6]
```

A measurable property: number of words.

Knowledge

In a study of political discourse: do critical comments tend to be longer?

→ a hypothesis about emotional charge or argumentation

Discussion: when is “knowledge” well founded?

“Negative comments are longer”

- Are three posts enough?
- Who decided which ones are “negative”?
- From which platform, over which period?

Knowledge comes from interpretation within a question: here it is a hypothesis to be tested.

Data sources

- Sensors and devices
- Information systems, databases
- Web platforms, social networks
- Digitised texts, archives, historical sources

Why the source matters

Provenance determines:

- quality
- completeness
- which methods we can apply

Forms of data

Structured

dept	age
CS	22
Math	21

tables, clear columns and types

Semi-structured

```
{"user": "u17",
 "text": "I support...",
 "likes": 12}
```

internal structure, no strict schema (JSON, XML)

Unstructured

“I do not trust the system”

text, images, audio: need extensive processing

All are “data”, but not alike

```
# Numerical (structured)
ages = [22, 25, 21, 30, 28]

# Categorical
departments = ["CS", "CS", "Math", "CS", "Physics"]

# Unstructured (linguistic)
texts = ["Data science is fascinating",
         "Data analysis requires critical thinking",
         "Language is inherently ambiguous"]
```

- A mean makes sense for ages, not for departments
- texts first need a **representation**

Language data: the focus of this book

Language is mostly unstructured data

- News, posts, political texts
- Educational material, medical reports
- They need **specialised techniques**

Existence ≠ usefulness

Data quality determines how reliable the results are

Four dimensions: **accuracy** · **completeness** · **consistency** · **timeliness**

Four dimensions of quality

Dimension	Question	Example in text
Accuracy	Does it faithfully capture the phenomenon?	errors in automatic transcription
Completeness	Is anything missing?	truncated sentences, no time or source
Consistency	Are there contradictions?	same text, different labels
Timeliness	Is it current enough?	neologisms missing from old data

Accuracy

- Do the data faithfully capture what we study?

In natural language

- Speech transcribed automatically, with errors
- Texts filed under the wrong category
- A “positive” label on a sarcastic comment

Completeness

- Are values, texts or metadata missing?

In natural language

- Fragmentary texts, incomplete sentences
- Words cut out by anonymisation
- Comments with no time or source → limited analysis

Consistency

- Contradictions within or across datasets
- Different labels for the same text
- Same phenomenon, coded differently

Objective or subjective tasks?

Objective: language of a text, objects in an image

Subjective: sentiment, toxicity

Intra-annotator disagreement (the same annotator changes their mind) is measured by repeating items.

Timeliness

Language changes over time

- New words, expressions, topics
- A model trained on old news → misses current references
- Result: lower performance

Exercise: find the quality problems

#	text	label	date
1	The new policy is unfair	negative	2024-03-02
2	Great, another delay...	positive	2024-03-02
3	I support the ref___	positive	—
4	The new policy is unfair	neutral	2024-03-05
5	A very good new measure	positive	2016-11-20

Hypothetical data. Which dimension does each row violate?

Exercise: answers

- **Row 2**: sarcasm labelled “positive” → accuracy
- **Row 3**: truncated word, no date → completeness
- **Rows 1 and 4**: same text, different label → consistency
- **Row 5**: predates the policy → timeliness

Row 2 is also a question of subjectivity: is it really an “error”?

Data are not neutral

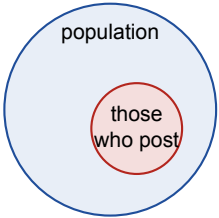
The outcome of human choices

- **Collection**: what gets recorded and what does not
- **Annotation**: who interprets, and how
- **Interpretation**: which questions, models, criteria

Subjectivity is not a defect: it is an inherent property that we manage consciously.

Subjectivity in collection

- Social networks ≠ the population
- Only those who choose to post publicly are heard
- News corpora reflect editorial practice and political lines



Our posts: **who did not write them?**

Subjectivity in annotation

```
text = "This measure is really interesting."
annotations = {
    "annotator_1": "positive",
    "annotator_2": "neutral",
    "annotator_3": "sarcastic",
}
```

- Same text, three readings
- Disagreement is **not an error**: it reflects the ambiguity of language

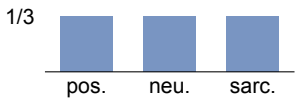
Model the disagreement, don't hide it

Majority vote

one "correct" label: here there is not even a majority

Label distribution

positive 1/3 · neutral 1/3 · sarcastic 1/3



Annotation, agreement and polarisation: chapters 12, 13 and 20.

Data do not “speak for themselves”

- Conclusions depend on:
  - the questions we ask
  - the algorithms we choose
  - the evaluation criteria

Example

A “toxicity” classifier trained on data that embed historical inequalities may reinforce stereotypes or silence social groups.

What this means in practice

Data = also a social construct

- A critical stance towards sources
- Transparency in collection and annotation
- Careful interpretation, especially when decisions are affected

Four roles

Role	Central question	Typical work
Data Analyst	What happened?	summaries, charts, reports
Data Scientist	What will happen? Why?	models, experiments, interpretation
ML Engineer	Does it run reliably?	integration, infrastructure, monitoring
Research Scientist	What new can we do?	new methods, publications

Data Analyst and Data Scientist

Data Analyst

- Descriptive analysis, visualisation
- Trends, patterns, deviations
- Translates findings for a non-technical audience

Data Scientist

- Prediction, explanation, generalisation
- ML models, experiments, comparisons
- Bridge: question $\leftrightarrow$ algorithm

ML Engineer and Research Scientist

Machine Learning Engineer

- From prototype to production system
- Infrastructure, data pipelines
- Monitors performance and stability over time

Research Scientist

- New methods, algorithms, theory
- Academia, research labs
- Long-term progress of knowledge

Same question, four perspectives

What do citizens say about the new policy?

Analyst: dashboard of comment volume and tone per week

Scientist: stance classifier, model comparison

ML engineer: the classifier runs daily, with performance checks

Researcher: a new method for sarcasm and annotator disagreement

The roles overlap and collaborate across the whole life cycle.

Example applications

Social networks and media

spread of (mis)information, communities, attitudes

Political discourse and public policy

parliamentary records, legislation, campaign speech

Recommender systems

personalisation from usage history and content

Health, education, finance

diagnosis, performance prediction, fraud detection

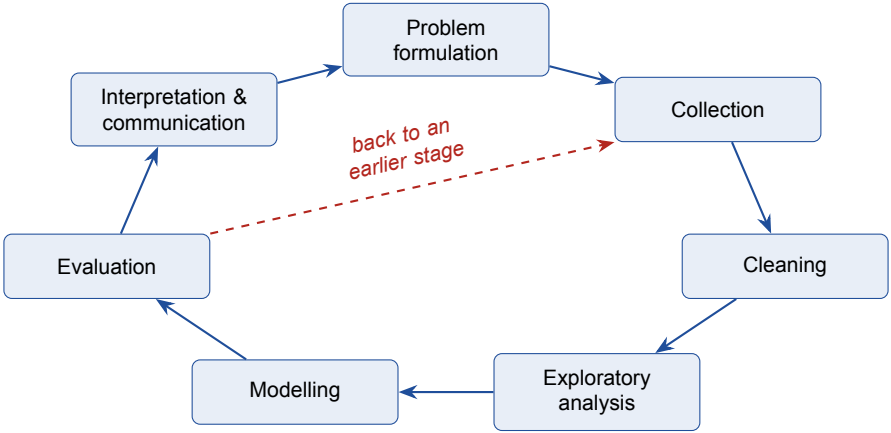
High-stakes decisions

Health · Education · Finance

Ethical, legal and social challenges require:

transparency · **interpretability** · **accountability**

The life cycle



1. Understanding and formulating the problem

- A research problem or an operational one?
- Question, **success criteria**, hypotheses, constraints
- The literature review starts here (and never stops)

Vague
"Analyse the comments on the policy."

Clear
"Does the share of negative comments change after the announcement? Success: a weekly estimate, with uncertainty."

2. Data collection

- Databases and data warehouses
- APIs
- Web crawling and scraping
- Experimental measurements

Scraping and APIs: chapters 8–11.

Not only technical

Are the data suitable? Is it **legal** to use them?

3. Cleaning and preprocessing

- Missing or wrong values, noise, outliers
- Merging sources, converting formats
- Time-consuming, but it shapes everything that follows

```
texts = ["The economy needs stability",
         "Education is a priority",
         "Citizens are demanding transparency"]
clean_texts = [t.lower() for t in texts]
```

Pitfall: lowercasing merges words

```
"US".lower()          # 'us'
"US".lower() == "us"   # True
```

- The country and the pronoun become **the same token**
- In Greek the opposite happens: capitals carry no accents, so one word ends up with **two forms**
- Every preprocessing step is a **decision** with consequences

Discussion: when should we not lowercase?

4. Exploratory data analysis

- Structure, distributions, relations
- Patterns and anomalies, testing hypotheses
- A bridge to choosing a model

```
lengths = [len(t.split()) for t in clean_texts]
average_length = sum(lengths) / len(lengths)
print(average_length)   # 4.0
```

A simple statistic, but one worth knowing before modelling.

Prediction · classification · understanding

- The choice of model depends on:
  - the nature of the problem
  - the type and quality of the data
  - requirements for interpretability and performance
- Essentially **experimental**: try, improve, repeat

6. Evaluation

Quantitative

- accuracy, error
- comparison with a simple baseline

Qualitative

- error analysis
- where does the model fail?

The central risk

Overfitting: good on training data, poor on new data. We want **generalisation**.

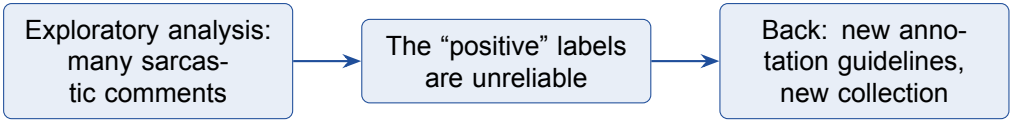
Model evaluation: chapter 17.

Accuracy is not enough

A project succeeds when it is:

- presented **clearly**, to technical or non-technical audiences
- tied back to the **original problem**
- able to support **decisions**

The cycle is iterative



- Findings in later stages → return to earlier ones
- Reformulate the problem, collect more data
- Non-linearity is a **feature**, not a failure

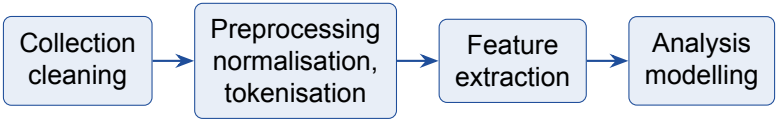
Why language is hard

Ambiguity	“I went to the *bank*”: river or money?
Context	“*Great*, another delay”: the rest changes the meaning
Subjectivity	“A really interesting measure”: positive or sarcastic?

Goal: from rich but shapeless material to representations we can compute with.

From text to numbers

A computer does not “read” meaning



The representation decides what analysis is possible, and how well it works.

Two families of representations

Statistical (frequencies)

- Count words
- Ignore syntax and meaning
- Basis of many early applications

Distributed (neural)

- Dense vectors
- Capture meaning and context
- Basis of modern models

Today we start with the first; the second comes in chapters 15–16.

Step 1: vocabulary

```
texts = ["data science",
         "datasets and language",
         "analysis of natural languages"]
vocabulary = sorted(set(" ".join(texts).split()))
```

```
['analysis', 'and', 'data', 'datasets', 'language',
 'languages', 'natural', 'of', 'science']
```

- set: each word once
- sorted: a fixed order, because each word becomes a **position** in the vector

Observation: word forms matter

9 “words”, but how many concepts?

data ≠ datasets

language ≠ languages

- Plurals and compounds create new “words”
- The vocabulary grows and similarities get lost

Much stronger in inflected languages like Greek. Fixes (lemmatisation, subwords) come later.

Step 2: Bag-of-Words

```
def vectorize(text, vocab):
    return [text.split().count(word) for word in vocab]

vectors = [vectorize(t, vocabulary) for t in texts]
```

- Every text → a vector of **fixed length**
- Every dimension = one vocabulary word
- Value = how many times it occurs
- Word order is **ignored**

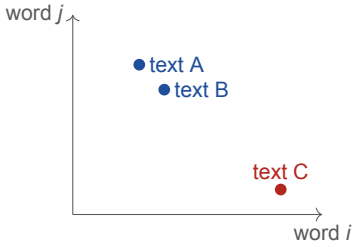
The document–term matrix

	analysis	and	data	datasets	language	languages	natural	of	science
data science	0	0	**1**	0	0	0	0	0	**1**
datasets and language	0	**1**	0	**1**	**1**	0	0	0	0
analysis of natural languages	**1**	0	0	0	0	**1**	**1**	**1**	0

Unstructured text → **structured data**, ready for statistics and ML

Interpretation: texts as points in space

- Every text = a point
- Nearby points = **shared words**
- Comparing texts = geometric distance
- The basis for clustering, classification, search



Question: which texts are similar?

*“**data** science”* vs *“**datasets** and language”*

Shared words in Bag-of-Words: 0

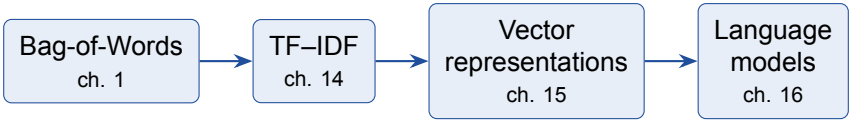
- Both are about data
- A different word form makes them look unrelated
- The representation decides what counts as “similar”

Limitations of Bag-of-Words

Order	“the child bit the dog” = “the dog bit the child”
Meaning	“car” and “automobile”: completely independent dimensions
Scale	large corpora → sparse, very high-dimensional vectors

Both sentences get the same vector: bit 1, child 1, dog 1, the 2.

What comes next



All extensions of the same idea: **text → vector**

Natural language processing tasks

- **Classification**: topic, sentiment, stance
- **Clustering** and topic analysis
- **Extraction** of entities, events, relations
- **Discourse analysis**: not only what is said, but how and why

NLP is not a separate field: it is data science on particularly complex data.

Social impact

It shapes

- how citizens are informed
- how institutions work
- access to knowledge

Risks

- bias
- lack of interpretability
- reproducing stereotypes

Understanding the limits is part of the analyst's **scientific responsibility**.

Summary

1. Data → information → knowledge, through organisation and interpretation
2. Quality has four dimensions and must always be checked
3. Data are not neutral; disagreement is information
4. A project is a cycle, not a line
5. Text becomes a vector; the representation defines similarity

Discussion questions

1. You want to measure attitudes to a policy from posts. **Who** does your sample leave out?
2. Three annotators disagree. Do you keep the majority or the distribution? **What changes** in evaluation?
3. Find two sentences with opposite meanings and the **same** Bag-of-Words vector.